QR Code for
report verification

Case ID	:	2600131590	Sample Type	:	BLOOD IN CCF DNA TUBE
Name	:	MRS. G. INDIRA (RPR2604140002)	Date & Time Collected	:	14-Apr-2026 12:00 AM
Sex/Age	:	Female/68 Years	Date & Time Received	:	18-Apr-2026 10:50 AM
Bill. Loc.	:	OncQuest Laboratories Ltd., New Delhi	Date & Time Reported	:	30-Apr-2026 05:56 PM
Ref. By	:		Report Version	:	1
Indication	:				

Liquidseq Lung Cancer Panel
On Illumina Novaseq 6000 Platform

Clinical Indication:

NA

Negative
(No clinically significant Variants detected)

Key Findings:

Tier	Gene & Transcript	Variant	Exon	Coverage/VAF
-Negative-				

Fusion Genes	Breakpoints	Total Reads
-Negative-		

CNV Type	Genes	Copy Number
-Negative-		

Biomarkers Results:

Assay Biomarker	Result	Interpretation
Microsatellite Instability (MSI)	Stable	Stable Microsatellite detected

Test Description

Liquidseq Lung Cancer Panel is advancing precision oncology research through the analysis of multiple relevant biomarkers in a single next-generation sequencing (NGS) test.

The Liquidseq Lung Cancer assay analyses **71** unique genes (**65** DNA + **6** Fusions driver genes) for single gene and multiple

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gene biomarker insights, detects single gene biomarkers from all variant types including **SNVs, indels, CNVs, known and novel fusions, and splice variants.** Analyze complex multi-gene biomarkers for mutational signatures, including microsatellite instability (**MSI**) for immunotherapy research.

The performance of the panel was verified using both commercially sourced reference controls and cell-free DNA (cfTNA)samples.

Test Methodology

Extraction & Library Preparation: Cell free Total Nucleic acid (cfTNA) extracted from the liquid biopsy sample is used for targeted gene capture using a custom designed hybrid capture panel. In process quality controls were determined for prepared library.

Sequencing: The libraries were sequenced at mean depth: **>25,000x** on **Illumina** Novaseq- next generation sequencing platform.

Data Analysis: Sequencing data are analyzed using a customized and validated bioinformatics pipeline capable of detecting various classes of genomic alterations, including single nucleotide variants (**SNVs**), **Indels**, copy number variations (**CNVs**) and **fusions.** Reportable genomic alterations and fusions are prioritized, classified, and reported based on AMP-ASCO-CAP guidelines and NCCN guidelines.

Limit of Detection (LOD): The LOD for SNVs and Short InDels is 0.1% Variant allele Frequency (VAF) and for Fusions is 05 supporting reads.

Notes

Treatment decisions should not be based solely on the results of this test. Findings must be interpreted in the context of the patient's complete clinical profile—including pathological, radiological, and family history—to guide diagnosis, prognosis, and therapeutic decisions.

Therapy-related information in this report is derived from multiple sources including FDA-approved drug data, NCCN guidelines, peer-reviewed literature, standard clinical databases, and biomarker evidence strength. However, therapy options listed may not be suitable or beneficial for all patients. This report aims to provide a summary of potentially effective treatments, therapies that may pose increased risks, and those that may have limited benefit, based on genomic alterations identified. Prognostic and diagnostic biomarkers relevant to the disease context may also be reported.

SN Genelab Pvt Ltd does not guarantee completeness, accuracy, or eligibility for trial enrollment. Physicians must verify eligibility criteria for specific trials independently.

Identification of a genomic alteration does not guarantee therapeutic efficacy or predict ineffectiveness. Final decisions on treatment should rest with the treating physician. The absence of reported therapeutic options does not imply ineffectiveness or lack of safety of any medication selected independently.

All variant classifications and related clinical information are based on the latest evidence from peer-reviewed publications, public clinical databases, and medical guidelines (e.g., NCCN, ASCO, AMP) available at the time of report generation. However, literature is continuously evolving, and classification may change with emerging evidence.

The assay is designed and validated for somatic variant detection, it does not differentiate between somatic and germline variants. If indicated, germline testing and genetic counselling are recommended.

Poor-quality cfDNA may fail QC thresholds at various stages such as cfDNA extraction, library preparation, or bioinformatics, and may result in assay failure or compromised results, such as low gene coverage or shallow variant depth.

Variants with lower allele frequencies may be reported if they are of strong or potential clinical relevance, or if previously detected in the same patient. False positives or negatives below assay sensitivity cannot be ruled out.

CNVs/gene amplifications reported, Confirmation using orthogonal methods such as DDPCR is recommended when clinically indicated.

Variants in non-coding regions or deep intronic sequences are not assessed in this assay.

This assay is under the scope of NABL as Liquidseq Comprehensive Panel.

DNA Gene List				
AKT1	ALK	APC	ATM	BARD1
BRAF	BRCA1	BRCA2	BRIP1	CDK12
CDKN2A	CHEK1	CHEK2	CTNNB1	EGFR
ERBB2	ERBB3	ESR1	FANCA	FANCD2
FANCL	FGFR3	FOXL2	GNA11	GNAQ
HRAS	IDH1	IDH2	KIT	KRAS
MET	MRE11A	NBN	NF1	NF2
NRAS	NTRK1	NTRK2	NTRK3	PALB2
PDGFRA	PIK3CA	POLD1	POLE	PPP2R2A
PTEN	RAD50	RAD51	RAD51B	RAD51C

RAD51D	RAD52	RAD54L	RAF1	RET
ROS1	SDHA	SDHB	SDHC	SDHD
STK11	TERT	TP53	VHL	XRCC2

RNA Gene List				
ALK	EML4	NTRK1	NTRK2	NTRK3
ROS1				

MSI	% of Unstable Loci
MSI - Stable	0-19%
MSI - Low / Intermediate	20-30%
MSI - High	>20%

Tier	Interpretation
Tier IA	Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to US FDA-approved therapies for a specific type of tumor or have been included in professional guidelines as therapeutic, diagnostic, and/or prognostic biomarkers for specific types of tumors.
Tier IB	Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to a therapy based on well-powered studies with consensus from experts in the field, or have diagnostic and/or prognostic significance of certain diseases based on well-powered studies with expert consensus.
Tier IIC	Variants of Potential Clinical Significance: biomarkers that predict response or resistance to therapies approved by FDA or professional societies for a different tumor type (i.e., off-label use of a drug), serve as inclusion criteria for clinical trials, or have diagnostic and/or prognostic significance based on the results of multiple small studies.
Tier IID	Variants of Potential Clinical Significance: Biomarkers that show plausible therapeutic significance based on preclinical studies or may assist disease diagnosis and/or prognosis themselves or along with other biomarkers based on small studies or multiple case reports with no consensus.
Tier III	Variants of Unknown clinical significance (VUS): Not observed at a significant allele frequency in the general or specific subpopulation or pancancer or tumor specific variant databases. No convincing published evidence of cancer Association

References:

1. *Exome Aggregation Consortium et al. "Analysis of Protein-Coding Genetic Variation in 60,706 Humans." Nature (2016) 536(7616): 285–291. PMID: 27535533.*

2. *Forbes, Simon A., et al. "COSMIC: somatic cancer genetics at high-resolution." Nucleic Acids Research (2016) 45(D1): D777-D783. PMID: 27899578.*

3. *Li, Marilyn M., et al. "Standards and guidelines for the interpretation and reporting of sequence variants in cancer: a joint consensus recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists." The Journal of Molecular Diagnostics (2017) 19(1): 4-23. PMID: 27993330.*

4. *Mamanova L, Coffey AJ, Scott CE, Kozarewa I, Turner EH, Kumar A, Howard E, Shendure J, Turner DJ: Target-enrichment strategies for next-generation sequencing. Nat Methods 2010, 7: 111–118*

5. *Richards, Sue, et al. "Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology." Genetics in Medicine (2015) 17(5): 405. PMID: 25741868.*

6. *The 1000 Genomes Project Consortium. "A Global Reference for Human Genetic Variation." Nature (2015) 526(7571): 68–74. PMID: 26432245.*

7. *Thusberg J, Olatubosun A, Vihinen M. Performance of mutation pathogenicity prediction methods on missense variants. Hum Mutat. 2011;32:358–368. doi: 10.1002/humu.21445.*

----------------- End Of Report -----------------

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